

v1.0 Antenna Build & Tuning Guide

Project Name: ARGUS

Antenna Type: *Tape-Measure Dipole*

Frequency Band: *UHF – 433 MHz*

Version: v1.0

Mounting & Deployment: Snap mounts, antenna wrapped around cubesat in stowed configuration

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1. Objective

This document provides detailed instructions for constructing, assembling, and tuning the primary communication antenna for the ARGUS CubeSat mission. The antenna is designed to operate in the 433 MHz UHF amateur radio band and is intended for deployment from the CubeSat bus and interfacing with a ground station receiver. This is the v1.0 of the antenna, new design iterations should be added as their own documentation and should follow this template if possible.

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2. Materials and Tools

Materials

- Spring steel tape, width ¼”
- U.FL to U.FL coaxial cable
- Fully assembled -Z board

Tools

- Calipers
- Soldering iron + solder
- Dremel with polishing/sanding wheel

- Oscilloscope or VNA (Vector Network Analyzer)

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3. Design Specifications

Key parameters:

- **Operating Frequency:** 433 MHz
- **Target VSWR:** < 1.10 (the closest to 1 the better)
- **Impedance Match:** 50 Ω
- **Polarization:** *Linear*
- **Deployment Method:** *Tape measure wrapped around the X's faces that self deploy with burn wire*
- **Radiation Pattern:** *Omnidirectional*
- **Gain (dBi):** 0
- **Length of Radiating Elements:** 17.3 cm (x2)

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4. Construction Instructions

Step-by-step guide to fabricate and assemble:

4.1 Cutting Radiating Elements

Measure and cut the radiating elements (2) to a length of 17.3 cm, corresponding to λ/4 of the target frequency.

4.2 Preparing & Assembling

Use the dremel with the polishing wheel to sand one end of each radiating element. (Otherwise the coating will melt when soldering to the board). Align the tips that have been sanded with the foot prints for the antenna on the -Z board and solder them. The curved side of the tape measure should be away from the board. Ensure solid mechanical support and strain relief. Slide the antenna mount (35deg angle) on each radiating element and snap them in place on the -Z board.

4.3 Mounting

The antenna board is now assembled and will be ready to assemble onto the CubeSat chassis once tuned (see 5.). After the antenna has been tuned, connect the U.FL to U.FL cable from the U.FL antenna port to the U.FL radio port on the mainboard. Secure the -Z board to the CubeSat chassis using 8x 6mm M3 hex screws. Ensure no contact with surrounding structures and clear deployment paths.

4.4 Deployment Test

Punch a 1/8” hole at the extremity of each radiating element using the hole punch tool. Cut 2x 30 cm pieces of the 20lb fishing line and tie each piece to each hole you just made. Wrap the antenna around the cubesat and tie the other end of the fishing line to the burn wire. Ensure the antenna is wrapped tightly and doesn’t extrude by more than 6mm from the -Z board. Cut the extra length of the fishing line. (Do NOT make a loop (passing the fishing line through the hole and burn wire and then tying) - this increases the risk for the fishing line to get stuck in the mounting hardware when deploying). Verify that the antenna reliably deploys from the stowed configuration.

** Add photos for every step**

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5. Tuning and Testing Procedures

5.1 Initial Check with VNA

Connect the antenna to the VNA port 1 using a U.FL to SMA cable. Sweep from at a center frequency of 433 MHz. (You may need to calibrate the VNA at the 433MHz frequency if it is not already - see below for how to calibrate the VNA). Record the S11 plot and note the resonant frequency.

5.2 Trim Adjustment

If the resonant frequency is above/below target, adjust the element length by ± 1/16” increments and retest until the peak resonant frequency matches the target frequency (433MHz).

5.3 SWR Measurement

Use an SWR meter or VNA to verify VSWR is below 1.10:1 at the operating frequency.

5.4 Radiation Pattern Testing

If access to an anechoic chamber is available, map the radiation pattern and confirm expected gain. (*See Anechoic Chamber Setup & Instructions for reference*)

** Add photos for every step**

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6. Troubleshooting Guide

Issue	Possible Cause	Fix
High SWR	Incorrect length	Trim or extend elements
Resonant frequency too low	Elements too long	Trim slightly and re-test

Poor radiation	Metal obstructions nearby	Check mounting location
No deployment	Mechanical jam	Re-test spring or burn wire

** Add VNA calibration steps & photos **